

Execution-order comparison

Pipeline: Normal LLM | 7 runs | 25 identical question texts | Reference: R4

Run	Folder / model
R1	a example-model
R2	run_2 example-model
R3	run_3 example-model
R4	run_4 example-model
R5	run_5 example-model
R6	run_6 example-model
R7	run_7 example-model

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R1, R2, R3, R4, R5, R6, R7, 25 g CO2e. Spread between highest and lowest: 0 g (0% relative to the lowest).

R1 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q3, 0 g; execution position 3 to 1. Net change over these matched successful pairs: 0 g.

R1: energy_wh change summed over 25 matched successful pairs: 0.

R1: latency_s change summed over 25 matched successful pairs: 0.

R2 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q3, 0 g; execution position 3 to 1. Net change over these matched successful pairs: 0 g.

R2: energy_wh change summed over 25 matched successful pairs: 0.

R2: latency_s change summed over 25 matched successful pairs: 0.

R3 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q1, 0 g; execution position 1 to 1. Net change over these matched successful pairs: 0 g.

R3: energy_wh change summed over 25 matched successful pairs: 0.

R3: latency_s change summed over 25 matched successful pairs: 0.

R5 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q24, 0 g; execution position 24 to 1. Net change over these matched successful pairs: 0 g.

R5: energy_wh change summed over 25 matched successful pairs: 0.

R5: latency_s change summed over 25 matched successful pairs: 0.

R6 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q24, 0 g; execution position 24 to 1. Net change over these matched successful pairs: 0 g.

R6: energy_wh change summed over 25 matched successful pairs: 0.

R6: latency_s change summed over 25 matched successful pairs: 0.

R7 versus R4: different execution sequence. 25 valid successful question pairs; 0 lower, 0 higher, 25 unchanged emissions.

Largest absolute change: Q22, 0 g; execution position 22 to 1. Net change over these matched successful pairs: 0 g.

R7: energy_wh change summed over 25 matched successful pairs: 0.

R7: latency_s change summed over 25 matched successful pairs: 0.

Recorded metadata differences: Ordering, Seed. Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Recorded shuffling schemes

These rules and seeds come from saved ordering metadata, with CSV metadata used when available. Missing rules are not inferred from the observed sequence. A shuffled scheme can sometimes produce the same sequence.

Run	Recorded scheme	Seed
R1	Keep original order	Not applicable
R2	Shuffle question types within each topic sheet; keep topic sheet order and question order within each question type	0
R3	Shuffle topic sheets, question types, and questions within each question type	0
R4 (reference)	Shuffle questions within each question type; keep topic sheets and question types in original order	0
R5	Keep topic sheet order; freely shuffle all questions within each topic sheet	0
R6	Freely shuffle all questions across all topic sheets	0
R7	Scheme not recorded	Not recorded

Which levels can move?

Run	Topic sheets	Question types	Individual questions	Group boundaries
R1	Original order	Original order	Original order	Groups stay together
R2	Original order	Shuffled within each topic	Original order within each question type	Groups stay together
R3	Shuffled	Shuffled within each topic	Shuffled within each question type	Groups stay together
R4	Original order	Original order	Shuffled within each question type	Groups stay together
R5	Original order	Can mix within each topic	Freely shuffled within each topic	Topic sheets stay together; question type groups can mix
R6	Can mix across topics	Can mix across question types	Freely shuffled across all selected questions	Topic sheets and question type groups can mix
R7	Not recorded	Not recorded	Not recorded	Not recorded

The rules apply only to the selected questions. A range may contain partial topic sheets or question type groups. Follow-Up questions form one question type group.

Execution movement summary

Positions are compared with R4. Earlier/later describes execution timing in the sequence, not an improvement or increase in emissions. The chosen reference can itself be shuffled.

Run	Questions moved	Same position	Largest position change	Observed sequence
R1	24	1	21	Different sequence
R2	25	0	23	Different sequence
R3	24	1	23	Different sequence
R4	0	25	0	Identical sequence
R5	24	1	23	Different sequence
R6	25	0	23	Different sequence
R7	24	1	21	Different sequence

Question label key

Q labels follow the execution order in R4. The same exact question text has the same label in every run and throughout this PDF.

Label	Exact question text
Q1	Question 03: Example question text for the sequence comparison.
Q2	Question 02: Example question text for the sequence comparison.
Q3	Question 01: Example question text for the sequence comparison.
Q4	Question 25: Example question text for the sequence comparison.
Q5	Question 24: Example question text for the sequence comparison.
Q6	Question 23: Example question text for the sequence comparison.
Q7	Question 22: Example question text for the sequence comparison.
Q8	Question 21: Example question text for the sequence comparison.
Q9	Question 20: Example question text for the sequence comparison.
Q10	Question 19: Example question text for the sequence comparison.
Q11	Question 18: Example question text for the sequence comparison.
Q12	Question 17: Example question text for the sequence comparison.
Q13	Question 16: Example question text for the sequence comparison.
Q14	Question 15: Example question text for the sequence comparison.
Q15	Question 14: Example question text for the sequence comparison.
Q16	Question 13: Example question text for the sequence comparison.
Q17	Question 12: Example question text for the sequence comparison.
Q18	Question 11: Example question text for the sequence comparison.
Q19	Question 10: Example question text for the sequence comparison.
Q20	Question 09: Example question text for the sequence comparison.
Q21	Question 08: Example question text for the sequence comparison.
Q22	Question 07: Example question text for the sequence comparison.
Q23	Question 06: Example question text for the sequence comparison.
Q24	Question 05: Example question text for the sequence comparison.
Q25	Question 04: Example question text for the sequence comparison.

Execution sequence comparison

Positions 1-12. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R1 Keep original order Seed: Not applicable	R2 Shuffle question types within each topic sheet; keep topic sheet order and question order within each question type Seed: 0
1	= Q1 Reference	↑ Q3 Earlier by 2	↑ Q3 Earlier by 2
2	= Q2 Reference	= Q2 Same position	↑ Q4 Earlier by 2
3	= Q3 Reference	↓ Q1 Later by 2	↑ Q5 Earlier by 2
4	= Q4 Reference	↑ Q25 Earlier by 21	↑ Q6 Earlier by 2
5	= Q5 Reference	↑ Q24 Earlier by 19	↑ Q7 Earlier by 2
6	= Q6 Reference	↑ Q23 Earlier by 17	↑ Q8 Earlier by 2
7	= Q7 Reference	↑ Q22 Earlier by 15	↑ Q9 Earlier by 2
8	= Q8 Reference	↑ Q21 Earlier by 13	↑ Q10 Earlier by 2
9	= Q9 Reference	↑ Q20 Earlier by 11	↑ Q11 Earlier by 2
10	= Q10 Reference	↑ Q19 Earlier by 9	↑ Q12 Earlier by 2
11	= Q11 Reference	↑ Q18 Earlier by 7	↑ Q13 Earlier by 2
12	= Q12 Reference	↑ Q17 Earlier by 5	↑ Q14 Earlier by 2

Execution sequence comparison

Positions 13-24. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R1 Keep original order Seed: Not applicable	R2 Shuffle question types within each topic sheet; keep topic sheet order and question order within each question type Seed: 0
13	= Q13 Reference	↑ Q16 Earlier by 3	↑ Q15 Earlier by 2
14	= Q14 Reference	↑ Q15 Earlier by 1	↑ Q16 Earlier by 2
15	= Q15 Reference	↓ Q14 Later by 1	↑ Q17 Earlier by 2
16	= Q16 Reference	↓ Q13 Later by 3	↑ Q18 Earlier by 2
17	= Q17 Reference	↓ Q12 Later by 5	↑ Q19 Earlier by 2
18	= Q18 Reference	↓ Q11 Later by 7	↑ Q20 Earlier by 2
19	= Q19 Reference	↓ Q10 Later by 9	↑ Q21 Earlier by 2
20	= Q20 Reference	↓ Q9 Later by 11	↑ Q22 Earlier by 2
21	= Q21 Reference	↓ Q8 Later by 13	↑ Q23 Earlier by 2
22	= Q22 Reference	↓ Q7 Later by 15	↑ Q24 Earlier by 2
23	= Q23 Reference	↓ Q6 Later by 17	↑ Q25 Earlier by 2
24	= Q24 Reference	↓ Q5 Later by 19	↓ Q1 Later by 23

Execution sequence comparison

Positions 25-25. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R1 Keep original order Seed: Not applicable	R2 Shuffle question types within each topic sheet; keep topic sheet order and question order within each question type Seed: 0
25	= Q25 Reference	↓ Q4 Later by 21	↓ Q2 Later by 23

Execution sequence comparison

Positions 1-12. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R3 Shuffle topic sheets, question types, and questions within each question type Seed: 0	R5 Keep topic sheet order; freely shuffle all questions within each topic sheet Seed: 0
1	= Q1 Reference	= Q1 Same position	↑ Q24 Earlier by 23
2	= Q2 Reference	↑ Q25 Earlier by 23	↑ Q23 Earlier by 21
3	= Q3 Reference	↑ Q24 Earlier by 21	↑ Q22 Earlier by 19
4	= Q4 Reference	↑ Q23 Earlier by 19	↑ Q21 Earlier by 17
5	= Q5 Reference	↑ Q22 Earlier by 17	↑ Q20 Earlier by 15
6	= Q6 Reference	↑ Q21 Earlier by 15	↑ Q19 Earlier by 13
7	= Q7 Reference	↑ Q20 Earlier by 13	↑ Q18 Earlier by 11
8	= Q8 Reference	↑ Q19 Earlier by 11	↑ Q17 Earlier by 9
9	= Q9 Reference	↑ Q18 Earlier by 9	↑ Q16 Earlier by 7
10	= Q10 Reference	↑ Q17 Earlier by 7	↑ Q15 Earlier by 5
11	= Q11 Reference	↑ Q16 Earlier by 5	↑ Q14 Earlier by 3
12	= Q12 Reference	↑ Q15 Earlier by 3	↑ Q13 Earlier by 1

Execution sequence comparison

Positions 13-24. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R3 Shuffle topic sheets, question types, and questions within each question type Seed: 0	R5 Keep topic sheet order; freely shuffle all questions within each topic sheet Seed: 0
13	= Q13 Reference	↑ Q14 Earlier by 1	↓ Q12 Later by 1
14	= Q14 Reference	↓ Q13 Later by 1	↓ Q11 Later by 3
15	= Q15 Reference	↓ Q12 Later by 3	↓ Q10 Later by 5
16	= Q16 Reference	↓ Q11 Later by 5	↓ Q9 Later by 7
17	= Q17 Reference	↓ Q10 Later by 7	↓ Q8 Later by 9
18	= Q18 Reference	↓ Q9 Later by 9	↓ Q7 Later by 11
19	= Q19 Reference	↓ Q8 Later by 11	↓ Q6 Later by 13
20	= Q20 Reference	↓ Q7 Later by 13	↓ Q5 Later by 15
21	= Q21 Reference	↓ Q6 Later by 15	↓ Q4 Later by 17
22	= Q22 Reference	↓ Q5 Later by 17	↓ Q3 Later by 19
23	= Q23 Reference	↓ Q4 Later by 19	↓ Q2 Later by 21
24	= Q24 Reference	↓ Q3 Later by 21	↓ Q1 Later by 23

Execution sequence comparison

Positions 25-25. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R3 Shuffle topic sheets, question types, and questions within each question type Seed: 0	R5 Keep topic sheet order; freely shuffle all questions within each topic sheet Seed: 0
25	= Q25 Reference	↓ Q2 Later by 23	= Q25 Same position

Execution sequence comparison

Positions 1-12. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R6 Freely shuffle all questions across all topic sheets Seed: 0	R7 Scheme not recorded Seed: Not recorded
1	= Q1 Reference	↑ Q24 Earlier by 23	↑ Q22 Earlier by 21
2	= Q2 Reference	↑ Q25 Earlier by 23	↑ Q21 Earlier by 19
3	= Q3 Reference	↓ Q1 Later by 2	↑ Q20 Earlier by 17
4	= Q4 Reference	↓ Q2 Later by 2	↑ Q19 Earlier by 15
5	= Q5 Reference	↓ Q3 Later by 2	↑ Q18 Earlier by 13
6	= Q6 Reference	↓ Q4 Later by 2	↑ Q17 Earlier by 11
7	= Q7 Reference	↓ Q5 Later by 2	↑ Q16 Earlier by 9
8	= Q8 Reference	↓ Q6 Later by 2	↑ Q15 Earlier by 7
9	= Q9 Reference	↓ Q7 Later by 2	↑ Q14 Earlier by 5
10	= Q10 Reference	↓ Q8 Later by 2	↑ Q13 Earlier by 3
11	= Q11 Reference	↓ Q9 Later by 2	↑ Q12 Earlier by 1
12	= Q12 Reference	↓ Q10 Later by 2	↓ Q11 Later by 1

Execution sequence comparison

Positions 13-24. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R6 Freely shuffle all questions across all topic sheets Seed: 0	R7 Scheme not recorded Seed: Not recorded
13	= Q13 Reference	↓ Q11 Later by 2	↓ Q10 Later by 3
14	= Q14 Reference	↓ Q12 Later by 2	↓ Q9 Later by 5
15	= Q15 Reference	↓ Q13 Later by 2	↓ Q8 Later by 7
16	= Q16 Reference	↓ Q14 Later by 2	↓ Q7 Later by 9
17	= Q17 Reference	↓ Q15 Later by 2	↓ Q6 Later by 11
18	= Q18 Reference	↓ Q16 Later by 2	↓ Q5 Later by 13
19	= Q19 Reference	↓ Q17 Later by 2	↓ Q4 Later by 15
20	= Q20 Reference	↓ Q18 Later by 2	↓ Q3 Later by 17
21	= Q21 Reference	↓ Q19 Later by 2	↓ Q2 Later by 19
22	= Q22 Reference	↓ Q20 Later by 2	↓ Q1 Later by 21
23	= Q23 Reference	↓ Q21 Later by 2	↑ Q25 Earlier by 2
24	= Q24 Reference	↓ Q22 Later by 2	= Q24 Same position

Execution sequence comparison

Positions 25-25. Reference: R4. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Execution position	R4 - Reference Shuffle questions within each question type; keep topic sheets and question types in original order Seed: 0	R6 Freely shuffle all questions across all topic sheets Seed: 0	R7 Scheme not recorded Seed: Not recorded
25	= Q25 Reference	↓ Q23 Later by 2	↓ Q23 Later by 2

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	25 [25/25]	25 [25/25]	325 [25/25]	0
R2	25 [25/25]	25 [25/25]	325 [25/25]	0
R3	25 [25/25]	25 [25/25]	325 [25/25]	0
R4	25 [25/25]	25 [25/25]	325 [25/25]	0
R5	25 [25/25]	25 [25/25]	325 [25/25]	0
R6	25 [25/25]	25 [25/25]	325 [25/25]	0
R7	25 [25/25]	25 [25/25]	325 [25/25]	0

Batch costs: R1-R6

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		25
	Valid 25/25	
R2		25
	Valid 25/25	
R3		25
	Valid 25/25	
R4		25
	Valid 25/25	
R5		25
	Valid 25/25	
R6		25
	Valid 25/25	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1		25
	Valid 25/25	
R2		25
	Valid 25/25	
R3		25
	Valid 25/25	
R4		25
	Valid 25/25	
R5		25
	Valid 25/25	
R6		25
	Valid 25/25	

Batch costs: R7-R7

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R7		25
	Valid 25/25	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R7		25
	Valid 25/25	

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2	R3
Folder *	a	run_2	run_3
Model(s)	example-model	example-model	example-model
Pipeline	Normal LLM	Normal LLM	Normal LLM
Ordering *	Keep original order	Shuffle question types within each topic sheet; keep topic sheet order and question order within each question type	Shuffle topic sheets, question types, and questions within each question type
Seed *	Not recorded	0	0
Questions	25	25	25
Runtime (s)	Not recorded	Not recorded	Not recorded
Completion evidence	No summary; completion unverified	No summary; completion unverified	No summary; completion unverified
codecarbon_version	Not recorded	Not recorded	Not recorded
os	Not recorded	Not recorded	Not recorded
effective_scope	Not recorded	Not recorded	Not recorded
measure_power_secs	Not recorded	Not recorded	Not recorded
pue	Not recorded	Not recorded	Not recorded
country_iso_code	Not recorded	Not recorded	Not recorded
config_profile	Not recorded	Not recorded	Not recorded
configured_settings	Not recorded	Not recorded	Not recorded
hardware	Not recorded	Not recorded	Not recorded
backends	Not recorded	Not recorded	Not recorded
hardware_evidence	Not recorded	Not recorded	Not recorded

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R4	R5	R6
Folder *	run_4	run_5	run_6
Model(s)	example-model	example-model	example-model
Pipeline	Normal LLM	Normal LLM	Normal LLM
Ordering *	Shuffle questions within each question type; keep topic sheets and question types in original order	Keep topic sheet order; freely shuffle all questions within each topic sheet	Freely shuffle all questions across all topic sheets
Seed *	0	0	0
Questions	25	25	25
Runtime (s)	Not recorded	Not recorded	Not recorded
Completion evidence	No summary; completion unverified	No summary; completion unverified	No summary; completion unverified
codecarbon_version	Not recorded	Not recorded	Not recorded
os	Not recorded	Not recorded	Not recorded
effective_scope	Not recorded	Not recorded	Not recorded
measure_power_secs	Not recorded	Not recorded	Not recorded
pue	Not recorded	Not recorded	Not recorded
country_iso_code	Not recorded	Not recorded	Not recorded
config_profile	Not recorded	Not recorded	Not recorded
configured_settings	Not recorded	Not recorded	Not recorded
hardware	Not recorded	Not recorded	Not recorded
backends	Not recorded	Not recorded	Not recorded
hardware_evidence	Not recorded	Not recorded	Not recorded

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R7
Folder *	run_7
Model(s)	example-model
Pipeline	Normal LLM
Ordering *	Not recorded
Seed *	Not recorded
Questions	25
Runtime (s)	Not recorded
Completion evidence	No summary; completion unverified
codecarbon_version	Not recorded
os	Not recorded
effective_scope	Not recorded
measure_power_secs	Not recorded
pue	Not recorded
country_iso_code	Not recorded
config_profile	Not recorded
configured_settings	Not recorded
hardware	Not recorded
backends	Not recorded
hardware_evidence	Not recorded

Question emissions change (%)

Relative to R4. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2	R3	R4	R5	R6
Q1	0	0	0	0	0	0
Q2	0	0	0	0	0	0
Q3	0	0	0	0	0	0
Q4	0	0	0	0	0	0
Q5	0	0	0	0	0	0
Q6	0	0	0	0	0	0
Q7	0	0	0	0	0	0
Q8	0	0	0	0	0	0
Q9	0	0	0	0	0	0
Q10	0	0	0	0	0	0
Q11	0	0	0	0	0	0
Q12	0	0	0	0	0	0
Q13	0	0	0	0	0	0
Q14	0	0	0	0	0	0
Q15	0	0	0	0	0	0
Q16	0	0	0	0	0	0
Q17	0	0	0	0	0	0
Q18	0	0	0	0	0	0
Q19	0	0	0	0	0	0
Q20	0	0	0	0	0	0
Q21	0	0	0	0	0	0
Q22	0	0	0	0	0	0

Question emissions change (%)

Relative to R4. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2	R3	R4	R5	R6
Q23	0	0	0	0	0	0
Q24	0	0	0	0	0	0
Q25	0	0	0	0	0	0

Question emissions change (%)

Relative to R4. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R7
Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0
Q13	0
Q14	0
Q15	0
Q16	0
Q17	0
Q18	0
Q19	0
Q20	0
Q21	0
Q22	0

Question emissions change (%)

Relative to R4. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R7
Q23	0
Q24	0
Q25	0

Largest changes: R1 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

Largest changes: R2 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

Largest changes: R3 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

Largest changes: R5 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

Largest changes: R6 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

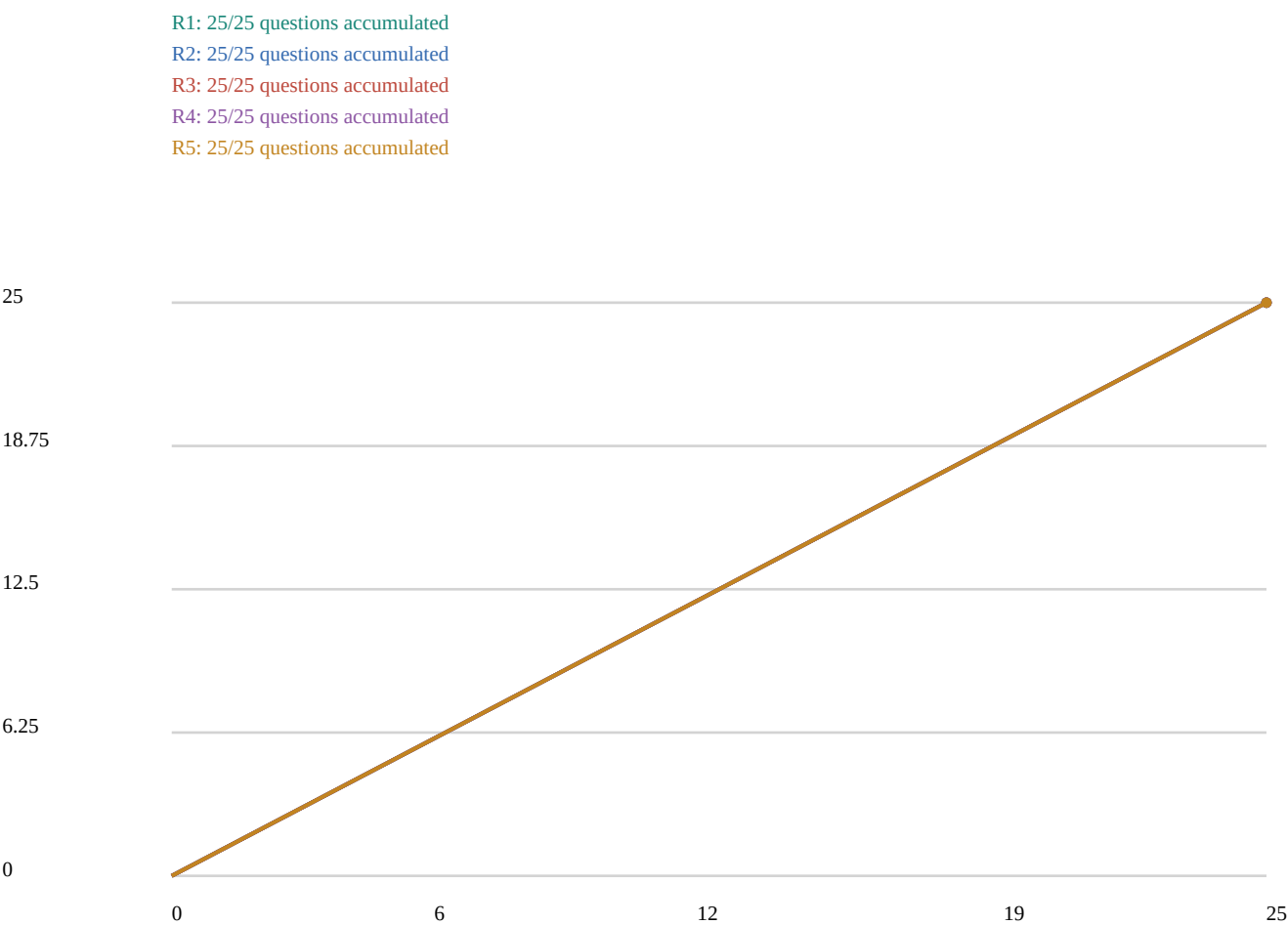
Largest changes: R7 versus R4

Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.

Q1	0
Q2	0
Q3	0
Q4	0
Q5	0
Q6	0
Q7	0
Q8	0
Q9	0
Q10	0
Q11	0
Q12	0

Cumulative emissions through execution

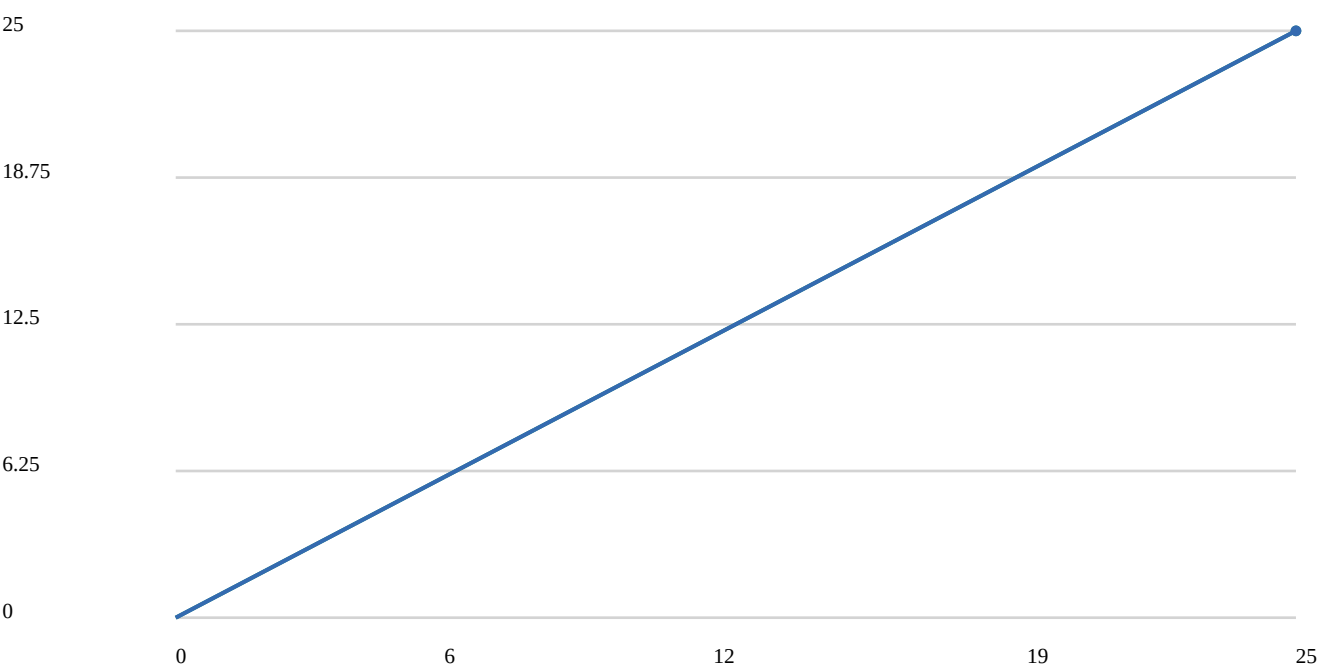
X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R6: 25/25 questions accumulated
R7: 25/25 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: Question 03: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	3	ok	1	1 (7/7)	1	3	0	0
R2	24	ok	1	1 (7/7)	1	24	0	0
R3	1	ok	1	1 (7/7)	1	1	0	0
R4	1	ok	1	1 (7/7)	1	1	0	0
R5	24	ok	1	1 (7/7)	1	24	0	0
R6	3	ok	1	1 (7/7)	1	3	0	0
R7	22	ok	1	1 (7/7)	1	22	0	0

Q2: Question 02: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	1	1 (7/7)	1	2	0	0
R2	25	ok	1	1 (7/7)	1	25	0	0
R3	25	ok	1	1 (7/7)	1	25	0	0
R4	2	ok	1	1 (7/7)	1	2	0	0
R5	23	ok	1	1 (7/7)	1	23	0	0
R6	4	ok	1	1 (7/7)	1	4	0	0
R7	21	ok	1	1 (7/7)	1	21	0	0

Q3: Question 01: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	1	1 (7/7)	1	1	0	0
R2	1	ok	1	1 (7/7)	1	1	0	0
R3	24	ok	1	1 (7/7)	1	24	0	0
R4	3	ok	1	1 (7/7)	1	3	0	0
R5	22	ok	1	1 (7/7)	1	22	0	0
R6	5	ok	1	1 (7/7)	1	5	0	0
R7	20	ok	1	1 (7/7)	1	20	0	0

Q4: Question 25: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	25	ok	1	1 (7/7)	1	25	0	0
R2	2	ok	1	1 (7/7)	1	2	0	0
R3	23	ok	1	1 (7/7)	1	23	0	0
R4	4	ok	1	1 (7/7)	1	4	0	0
R5	21	ok	1	1 (7/7)	1	21	0	0
R6	6	ok	1	1 (7/7)	1	6	0	0
R7	19	ok	1	1 (7/7)	1	19	0	0

Q5: Question 24: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	24	ok	1	1 (7/7)	1	24	0	0
R2	3	ok	1	1 (7/7)	1	3	0	0
R3	22	ok	1	1 (7/7)	1	22	0	0
R4	5	ok	1	1 (7/7)	1	5	0	0
R5	20	ok	1	1 (7/7)	1	20	0	0
R6	7	ok	1	1 (7/7)	1	7	0	0
R7	18	ok	1	1 (7/7)	1	18	0	0

Q6: Question 23: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	23	ok	1	1 (7/7)	1	23	0	0
R2	4	ok	1	1 (7/7)	1	4	0	0
R3	21	ok	1	1 (7/7)	1	21	0	0
R4	6	ok	1	1 (7/7)	1	6	0	0
R5	19	ok	1	1 (7/7)	1	19	0	0
R6	8	ok	1	1 (7/7)	1	8	0	0
R7	17	ok	1	1 (7/7)	1	17	0	0

Q7: Question 22: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	22	ok	1	1 (7/7)	1	22	0	0
R2	5	ok	1	1 (7/7)	1	5	0	0
R3	20	ok	1	1 (7/7)	1	20	0	0
R4	7	ok	1	1 (7/7)	1	7	0	0
R5	18	ok	1	1 (7/7)	1	18	0	0
R6	9	ok	1	1 (7/7)	1	9	0	0
R7	16	ok	1	1 (7/7)	1	16	0	0

Q8: Question 21: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	21	ok	1	1 (7/7)	1	21	0	0
R2	6	ok	1	1 (7/7)	1	6	0	0
R3	19	ok	1	1 (7/7)	1	19	0	0
R4	8	ok	1	1 (7/7)	1	8	0	0
R5	17	ok	1	1 (7/7)	1	17	0	0
R6	10	ok	1	1 (7/7)	1	10	0	0
R7	15	ok	1	1 (7/7)	1	15	0	0

Q9: Question 20: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	20	ok	1	1 (7/7)	1	20	0	0
R2	7	ok	1	1 (7/7)	1	7	0	0
R3	18	ok	1	1 (7/7)	1	18	0	0
R4	9	ok	1	1 (7/7)	1	9	0	0
R5	16	ok	1	1 (7/7)	1	16	0	0
R6	11	ok	1	1 (7/7)	1	11	0	0
R7	14	ok	1	1 (7/7)	1	14	0	0

Q10: Question 19: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	19	ok	1	1 (7/7)	1	19	0	0
R2	8	ok	1	1 (7/7)	1	8	0	0

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R3	17	ok	1	1 (7/7)	1	17	0	0
R4	10	ok	1	1 (7/7)	1	10	0	0
R5	15	ok	1	1 (7/7)	1	15	0	0
R6	12	ok	1	1 (7/7)	1	12	0	0
R7	13	ok	1	1 (7/7)	1	13	0	0

Q11: Question 18: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	18	ok	1	1 (7/7)	1	18	0	0
R2	9	ok	1	1 (7/7)	1	9	0	0
R3	16	ok	1	1 (7/7)	1	16	0	0
R4	11	ok	1	1 (7/7)	1	11	0	0
R5	14	ok	1	1 (7/7)	1	14	0	0
R6	13	ok	1	1 (7/7)	1	13	0	0
R7	12	ok	1	1 (7/7)	1	12	0	0

Q12: Question 17: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	17	ok	1	1 (7/7)	1	17	0	0
R2	10	ok	1	1 (7/7)	1	10	0	0
R3	15	ok	1	1 (7/7)	1	15	0	0
R4	12	ok	1	1 (7/7)	1	12	0	0
R5	13	ok	1	1 (7/7)	1	13	0	0
R6	14	ok	1	1 (7/7)	1	14	0	0
R7	11	ok	1	1 (7/7)	1	11	0	0

Q13: Question 16: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	16	ok	1	1 (7/7)	1	16	0	0
R2	11	ok	1	1 (7/7)	1	11	0	0
R3	14	ok	1	1 (7/7)	1	14	0	0
R4	13	ok	1	1 (7/7)	1	13	0	0

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R5	12	ok	1	1 (7/7)	1	12	0	0
R6	15	ok	1	1 (7/7)	1	15	0	0
R7	10	ok	1	1 (7/7)	1	10	0	0

Q14: Question 15: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	15	ok	1	1 (7/7)	1	15	0	0
R2	12	ok	1	1 (7/7)	1	12	0	0
R3	13	ok	1	1 (7/7)	1	13	0	0
R4	14	ok	1	1 (7/7)	1	14	0	0
R5	11	ok	1	1 (7/7)	1	11	0	0
R6	16	ok	1	1 (7/7)	1	16	0	0
R7	9	ok	1	1 (7/7)	1	9	0	0

Q15: Question 14: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	14	ok	1	1 (7/7)	1	14	0	0
R2	13	ok	1	1 (7/7)	1	13	0	0
R3	12	ok	1	1 (7/7)	1	12	0	0
R4	15	ok	1	1 (7/7)	1	15	0	0
R5	10	ok	1	1 (7/7)	1	10	0	0
R6	17	ok	1	1 (7/7)	1	17	0	0
R7	8	ok	1	1 (7/7)	1	8	0	0

Q16: Question 13: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	13	ok	1	1 (7/7)	1	13	0	0
R2	14	ok	1	1 (7/7)	1	14	0	0
R3	11	ok	1	1 (7/7)	1	11	0	0
R4	16	ok	1	1 (7/7)	1	16	0	0
R5	9	ok	1	1 (7/7)	1	9	0	0
R6	18	ok	1	1 (7/7)	1	18	0	0

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R7	7	ok	1	1 (7/7)	1	7	0	0

Q17: Question 12: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	12	ok	1	1 (7/7)	1	12	0	0
R2	15	ok	1	1 (7/7)	1	15	0	0
R3	10	ok	1	1 (7/7)	1	10	0	0
R4	17	ok	1	1 (7/7)	1	17	0	0
R5	8	ok	1	1 (7/7)	1	8	0	0
R6	19	ok	1	1 (7/7)	1	19	0	0
R7	6	ok	1	1 (7/7)	1	6	0	0

Q18: Question 11: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	11	ok	1	1 (7/7)	1	11	0	0
R2	16	ok	1	1 (7/7)	1	16	0	0
R3	9	ok	1	1 (7/7)	1	9	0	0
R4	18	ok	1	1 (7/7)	1	18	0	0
R5	7	ok	1	1 (7/7)	1	7	0	0
R6	20	ok	1	1 (7/7)	1	20	0	0
R7	5	ok	1	1 (7/7)	1	5	0	0

Q19: Question 10: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	10	ok	1	1 (7/7)	1	10	0	0
R2	17	ok	1	1 (7/7)	1	17	0	0
R3	8	ok	1	1 (7/7)	1	8	0	0
R4	19	ok	1	1 (7/7)	1	19	0	0
R5	6	ok	1	1 (7/7)	1	6	0	0
R6	21	ok	1	1 (7/7)	1	21	0	0
R7	4	ok	1	1 (7/7)	1	4	0	0

Q20: Question 09: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	9	ok	1	1 (7/7)	1	9	0	0
R2	18	ok	1	1 (7/7)	1	18	0	0
R3	7	ok	1	1 (7/7)	1	7	0	0
R4	20	ok	1	1 (7/7)	1	20	0	0
R5	5	ok	1	1 (7/7)	1	5	0	0
R6	22	ok	1	1 (7/7)	1	22	0	0
R7	3	ok	1	1 (7/7)	1	3	0	0

Q21: Question 08: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	8	ok	1	1 (7/7)	1	8	0	0
R2	19	ok	1	1 (7/7)	1	19	0	0
R3	6	ok	1	1 (7/7)	1	6	0	0
R4	21	ok	1	1 (7/7)	1	21	0	0
R5	4	ok	1	1 (7/7)	1	4	0	0
R6	23	ok	1	1 (7/7)	1	23	0	0
R7	2	ok	1	1 (7/7)	1	2	0	0

Q22: Question 07: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	7	ok	1	1 (7/7)	1	7	0	0
R2	20	ok	1	1 (7/7)	1	20	0	0
R3	5	ok	1	1 (7/7)	1	5	0	0
R4	22	ok	1	1 (7/7)	1	22	0	0
R5	3	ok	1	1 (7/7)	1	3	0	0
R6	24	ok	1	1 (7/7)	1	24	0	0
R7	1	ok	1	1 (7/7)	1	1	0	0

Q23: Question 06: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	6	ok	1	1 (7/7)	1	6	0	0

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R2	21	ok	1	1 (7/7)	1	21	0	0
R3	4	ok	1	1 (7/7)	1	4	0	0
R4	23	ok	1	1 (7/7)	1	23	0	0
R5	2	ok	1	1 (7/7)	1	2	0	0
R6	25	ok	1	1 (7/7)	1	25	0	0
R7	25	ok	1	1 (7/7)	1	25	0	0

Q24: Question 05: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	5	ok	1	1 (7/7)	1	5	0	0
R2	22	ok	1	1 (7/7)	1	22	0	0
R3	3	ok	1	1 (7/7)	1	3	0	0
R4	24	ok	1	1 (7/7)	1	24	0	0
R5	1	ok	1	1 (7/7)	1	1	0	0
R6	1	ok	1	1 (7/7)	1	1	0	0
R7	24	ok	1	1 (7/7)	1	24	0	0

Q25: Question 04: Example question text for the sequence comparison.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	4	ok	1	1 (7/7)	1	4	0	0
R2	23	ok	1	1 (7/7)	1	23	0	0
R3	2	ok	1	1 (7/7)	1	2	0	0
R4	25	ok	1	1 (7/7)	1	25	0	0
R5	25	ok	1	1 (7/7)	1	25	0	0
R6	2	ok	1	1 (7/7)	1	2	0	0
R7	23	ok	1	1 (7/7)	1	23	0	0